

Orogeny in Japan driven by nearby double subduction

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Abstract

Same-dip double subduction (SDDS) systems influence plate kinematics, geometry, and mantle flow within the region bounded by the two subduction zones. However, whether this influence extends to nearby plate margin tectonics remains unclear. To address this, we examine the tectonics of Northeast Japan in the context of the Ryukyu/Izu-Bonin-Marianas SDDS evolution and explore a potential geodynamic connection through 3-D geodynamic models. Our results reveal that this SDDS system, dragging the Pacific trench westward, drives a northward-propagating trench advance and compression impacting the Northeastern Japan arc and backarc. Thus, the dynamics of the Ryukyu/Izu-Bonin-Marianas SDDS over the past 10-5 million years (Myr) ago explains the enigmatic plate kinematics responsible for non-collisional orogeny and backarc subduction initiation in Northeast Japan since

~6-3.5 Myr ago. The orogenic effects of SDDS introduce a novel mechanism by which subduction zones establish the critical plate kinematic conditions for non-collisional mountain building. These findings provide essential insights into the seismotectonic behavior of Northeast Japan, a region prone to catastrophic earthquakes. Additionally, they improve our understanding of past orogenic processes and its connection to other proposed ancient SDDS systems.

Subduction zones are critical features of our planet, providing the forces needed to move tectonic plates and deform margins, develop volcanic arcs and sedimentary basins, and unleashing the most devastating megathrust earthquakes¹. In recent years, researchers have observed that these geodynamic settings can be significantly altered by the emergence of a nearby subduction zone, usually forming so-called double subduction systems. These impacts include dramatic changes in plate kinematics, trench motions, slab dips, and the associated mantle flow²⁻⁹.

Double subduction zones occur in three discernible geometries: divergent double subduction¹⁰, evident in active settings such as the Molucca Sea subduction zones⁹ and the Mediterranean region⁴; convergent double subduction, typified by areas hosting oppositely dipping subduction zones, such as those surrounding the Caribbean Plate¹¹ and Southeast Asia¹²; and same-dip double subduction (SDDS), which feature both ancient and active examples like the Mesozoic Neothetian arc system^{3,6} and the late Cenozoic Ryukyu/Izu-Bonin-Marianas SDDS^{2,5,13} (Fig. 1a). Notably, recent investigations have highlighted the significant geodynamic impact of SDDS initiation, elucidating its role in driving major changes in direction and/or plate velocity^{3,6,14}. Studies focused on the Ryukyu/Izu-Bonin-Marianas SDDS, involving the subduction of the Pacific Plate beneath the Philippine Sea Plate, which in turn subducts beneath

the Eurasian Plate along the Ryukyu-Nankai Trench, indicate that this system leads to Pacific slab steepening and trench advance^{2,5,13,15} (Fig. 1a-c). Taken together, these new findings suggest that the geodynamics of many subduction zones is best understood when considering these plate boundaries as components within larger-scale subduction networks. This opens up exciting new avenues for understanding active margin dynamics and, more specifically, the magnitudes and length scales at which double subduction is significant. In this regard, while most studies on SDDS have focused on the effects confined to the area of double plate convergence, the broader influence of SDDS on nearby plate margin tectonics remains unknown.

Here, we assess this issue by analyzing the geological evolution of Northeast Japan and the neighboring Ryukyu/Izu-Bonin-Marianas SDDS, presenting evidence for what we define as double subduction-induced orogeny. We hypothesize that the initiation of the Ryukyu/Izu-Bonin-Marianas SDDS modifies trench kinematics along the Pacific subduction, triggering non-collisional compression in Northeast Japan and inducing the active subduction initiation process in the eastern Japan Sea. We test this hypothesis through geologically constrained 3-D geodynamic numerical experiments on double subduction, illustrating how SDDS impacts tectonic stress within the overriding plates of adjacent active margins. The tectonic consequences of SDDS onset illustrate a novel mechanism whereby complex plate tectonic interactions create optimal plate kinematic conditions for non-collisional orogeny¹⁶.

Late Cenozoic tectonic evolution of Northeast Japan

During the Oligocene-Miocene (~25-20 Myr ago) several marginal basins, including the Japan Sea backarc basin, and the Shikoku and Parece Vela Basins

of the Philippine Sea Plate, underwent simultaneous opening caused by the eastward Pacific trench retreat and major strike slip tectonics^{17–20} (Figs. 1a and 2a,b). This process caused the Japanese Island arc to separate from the Asian continental margin through accelerated backarc rifting during the Early to Middle Miocene period^{17,20–22}. This culminated at ~16–15 Myr ago, resulting in the formation of a 3–3.7-kilometer-deep basin with rifted continental crust and local oceanic floor (e.g.,^{17,18,23,24}) (Figs. 1a and 2b). In Northeast Japan, neutral stress conditions during thermal subsidence progressively changed from mild east-west compression since ~10–8 Myr ago to highly compressional since ~5.6–3.5 Myr ago^{17–20,25,26} (Fig. 2b). Pliocene-Quaternary east-west compression led to 10–15 km of shortening within Northeast Japan²⁵, propagating westwards to the Japan Sea^{25,27} and the East Korean margin²⁸. Currently, the shortening is distributed across several east- and west-dipping thrusts and basin inversion structures^{19,20,25–27,29}, with rates reaching up to 15 mm/yr off Central Japan³⁰ (Fig. 1b). A compilation map of active faulting shows that compression affects a ~1,200 km-long deformed belt extending into the backarc region west of Hokkaido Island (see Fig. 5 in²⁰). The structural observations, coupled with the documentation of thrust earthquakes occurring in this area over the last century, with magnitudes ranging from Mw 6.9 to 7.8, have prompted several authors to propose incipient subduction initiation in the Eastern Japan Sea (e.g.,^{27,31,32}) (Fig. 1a,b). Active backarc faulting in this region was responsible for the recent Noto-Hanto earthquake (2024, Mw 7.6) that generated strong ground motion, large crustal deformation and tsunamis that caused significant damage in the region³³. Recent studies also reported geological and geophysical evidence for diffuse subduction initiation along the eastern Korean margin active since ~5 Myr ago²⁸ (Fig. 1a,b). To the south, tectonics are locally dominated by the interaction of the Izu-Bonin arc (Fig.

1a). This area is characterized by a main phase of arc subduction and deformation occurring since $\sim 6\text{--}5$ Myr ago and crustal motion within the active fault system controlled by the indented configuration of the plate boundary (e.g., ^{20,34}).

Competing Hypotheses for the Origin of Northeast Japan Compression

The origin of the Pliocene-Quaternary compressional tectonic regime across the Northeastern Japanese active margin and the backarc Japan Sea remains highly contentious. One hypothesis suggests a predominant role for the lower plate³⁵, proposing that the Pacific plate accelerated since ~ 5 Myr ago, introducing a significant compression component normal to the Japan trench³⁶. However, in this mechanism, it is unclear why compression localized only in Northeast Japan and not along the entire high-convergence western Pacific subduction³⁷. Moreover, in global surveys of all major subduction zones, high plate convergence does not correlate with upper plate compression³⁸. An alternative, classic hypothesis highlights the impact of eastward upper plate motion, which is attributed to the clockwise rotation of the Amurian plate in response to the distant effects of the India-Eurasia collision^{19,22,27}. This rotation would have triggered slow convergence between the Amurian Plate and the Okhotsk Plate along the eastern margin of the Japan Sea. Nevertheless, Amurian plate extrusion has been an ongoing process since at least the late Oligocene-Early Miocene³⁹, complicating a direct link with the Pliocene onset of compression in northeast Japan and its backarc region. Furthermore, while the motion of the upper plate towards the trench is significant, it alone does not suffice to initiate compression because upper plate deformation is driven by the relative difference between upper plate and trench motion¹⁶. It is therefore crucial to

examine how the movement of the upper plate interacts with trench dynamics to fully understand the triggering of compression.

Both global parametric analyses and simple conceptual/kinematic models indicate that plate margin compression is favored when the trench retreat velocity is less than the upper plate velocity³⁷. In the study area, the Japan Trench east of Northeast Japan is advancing at rates of 0.7 to 0.6 cm/yr in a seafloor-spreading aligned absolute reference frame⁴⁰, resulting in an overall convergence with the near-stationary upper plate (in the same reference frame), and thus meeting the plate kinematic conditions for upper plate compression (Fig. 1a). These kinematic conditions are still met even when analyzing other widely used absolute plate motion reference frames^{41–43}. Based on this, it has been suggested that the presence of a wide Pacific slab, preventing rapid slab rollback and trench retreat combined with a trenchward-moving upper plate, created this plate kinematic condition in Northeast Japan³⁷ (Fig. 1a,b). However, this conflicts with robust plate reconstructions including mantle slab records suggesting the subduction of a wide Pacific plate since 50–40 Myr ago (e.g.,⁴⁴), well before the onset of regional compression in the study area. The Pacific kinematic conditions driving compression in the study area have also been associated with the clockwise rotation and northwestward movement of the Philippine Sea Plate⁴⁵. However, major rotation of this plate occurred since at least ~30–40 Myr ago^{44,46}, again significantly before the onset of compression in Northeast Japan (Fig. 2a,b).

Alternatively, recent numerical experiments on Pacific subduction⁴⁷ suggest that the plate kinematics leading to compression may result from a combination of an imposed eastward asthenospheric flow and slab folding processes at depth, resulting in several trench retreat-advance cycles since 50 Myr ago. The issue with the applicability of this model to Northeast Japan

is that these experiments predict an extensional phase from 6 to 3 Myr ago, whereas geological data indicate strong and continuous compression since then^{17–20,25,26} (Fig. 2b). Moreover, the many alternations of compression and extension since ~35 Myr ago, as shown in their modeling results, do not correlate well with geological data and the overall tectonic evolution of this specific region (see previous section) (Fig. 2b).

Lastly, alternatives such as compression triggered by subduction or accretion of bathymetric highs, are unlikely due to the localized nature of the associated compression^{37,38}. More importantly, the absence of a major aseismic ridge or oceanic plateaus interacting with the northeastern Japan Trench—features necessary to induce eastward-directed compression—further diminishes the plausibility of this hypothesis. The only bathymetric high that may influence compression is the Izu-Bonin Arc (Fig. 1a). However, as previously shown³⁴, it is considered more of a local factor, with its main deformational effects not extending more than approximately 150–200 km north of the accretion zone. Therefore, it cannot account for the documented ~1,200 km-long region under compression, as evidenced by both active and ancient thrust faults, as well as thrust-related intraplate earthquakes observed over the past few decades (see Fig. 5 in Taira²⁰).

In summary, over the past ~5–6 Myr ago, the Northeast Japan arc-to-backarc region has experienced increasing compression due to the onset of a plate kinematic context associated with an advancing trench. Existing explanations for this phenomenon are either unfeasible^{35,36,47} or appear incomplete^{19,22,27,37,38,45}. More importantly, numerical geodynamic models specifically designed to test conceptual frameworks in this region are still lacking.

Double subduction-induced orogeny in Northeast Japan?

To investigate the geodynamic mechanisms driving effective plate kinematic conditions for non-collisional orogeny in Northeast Japan over the past ~ 5.6 -3 Myr ago, it is essential to consider trench kinematics and the Cenozoic evolution of Pacific subduction. Present-day trench kinematics⁴⁰, spanning from Northeast Japan to the Izu-Bonin-Marianas subduction segments (i.e., Pacific trench in this study), depict a southward increase in westward trench advance (Fig. 1a). This southward shift in trench motion, towards increased advance, is also observed in various trench motion catalogs and is independent of the chosen absolute reference frame^{41–43,48}. A potential explanation for the induced compression in Japan is then that forces associated with this Izu-Bonin-Marianas trench advance are transmitted northward to this region. In this scenario, the slowly moving upper plate resists the west-directed force associated with trench advance, as transmitted from the south, and that this results in compression in the Northeastern Japan arc and backarc region (Fig. 1a,b). If this is correct, we would expect a temporal correlation between the initiation of trench advance in the Izu-Bonin-Marianas subduction zone and compression in Northeastern Japan and the Japan Sea.

The onset of trench advance along the Izu-Bonin-Marianas subduction zone has been linked to a reorganization of major tectonic boundaries associated with the evolution of the Ryukyu/Izu-Bonin-Marianas SDDS^{2,5,13,15}. Between ~ 40 and 15 Myr ago, the Izu-Bonin-Marianas trench predominantly underwent rapid, intermittent retreat^{17–20}, shifting progressively to trench advance since ~ 10 –5 Myr ago^{2,5,13,15,49,50} (Fig. 2a,b). Numerical models support the hypothesis that subduction reactivation of the Philippine Sea Plate since 15-10 Myr ago in the Ryukyu-Nankai trench, following a slab detachment caused by

ridge subduction^{5,51}, exerted a pronounced northwestward pull on the Philippine Sea Plate, consequently driving Pacific trench advance and facilitating the steepening of the Pacific slab (Fig. 2a). Recent numerical studies indicate that the initial stage of Ryukyu subduction (~ 30 –15 Myr ago) did not significantly affect the Izu-Bonin-Marianas trench, most likely due to ineffective stress transmission caused by a thermally weakened upper plate at the Shikoku-Parece-Vela Basins backarc spreading center¹⁵. However, after this system cooled and strengthened/fused since ~ 15 –10 Myr ago, the reactivated Ryukyu subduction exerted a strong pull, driving the trench advance observed today^{5,52} (Fig. 2b).

We note that the significant shift in trench motion for the Izu-Bonin-Marianas trench since ~ 10 –5 Myr ago^{5,49,50,52} slightly precedes and coincides with the onset of strong compressional conditions in Northeast Japan and the Japan Sea backarc basin since ~ 5.6 –3 Myr ago^{17–20,25–28,32} (Fig. 2b). These observations indicate a potential causal relationship between the reactivation of the Ryukyu/Izu-Bonin-Marianas SDDS after Philippine Sea plate fusion/strengthening, trench advance, and the onset of the compressional regime in Northeast Japan and the incipient subduction in the Japan Sea.

3-D geodynamic models of double subduction-induced orogeny

To evaluate the viability of double subduction-induced orogeny in Northeast Japan, we set up 3-D numerical subduction models (see Methods; Fig. 3a–g, Supplementary Figs. 1–6). Our aim was to examine whether the insertion of an additional (Ryukyu-Nankai) subduction zone significantly enhances compression within the Northeast Japan upper plate region relative to a single

subduction model containing only Pacific plate subduction along the Japan-Izu-Bonin-Mariana trenches. We therefore modeled two subduction scenarios — with and without subduction along the Ryukyu-Nankai trench (i.e., single vs. double subduction) — using the ASPECT finite element code^{53–56} (Fig. 3a,b).

In our models, we solve for the velocity and stress fields associated with imposed slab and tectonic plate geometries. The plan-view margin shapes are an idealized representation of the present-day plate geometry. The Pacific and Philippine Sea subduction geometries are based on the present-day slabs, as delineated by earthquake hypocenters and seismic tomography^{57–59} (Fig. 1c). We do not model the transition from single to double subduction; rather, we intend to represent the change in stress state from before (~ 10 Myr ago) to significantly after Ryukyu-Nankai subduction reinitiation^{5,51} (present day). We therefore run our “pseudo-instantaneous” single and double subduction models for only 10 kyrs; this is sufficiently short that the plate and slab geometry undergoes minimal evolution from the geometries imposed. In both models, we considered a single, mechanically continuous (i.e., no weak backarc area¹⁵) upper plate, and identical Pacific plate/slab geometries. We adopt a linear viscous rheology for the lithosphere and crust, and we use a simplified power-law viscous rheology for the underlying flowing mantle. This enables us to hone in on the as yet unexplored first-order impact of double subduction on the upper plate stress state, while avoiding complexities associated with a large number of rheological parameters. Here we focus on two models (single and double subduction) with equivalent crustal viscosities ($\eta_{crust} = 5 \times 10^{19}$ Pas) and lithospheric strengths relative to the underlying mantle ($\eta_{lith.}/\eta_{mantle} = 500$). We chose the viscosity values to be intermediate within the ranges inferred from various geophysical, geodynamic, and experimental constraints (e.g.,^{60–65}),

and to produce tectonic plate velocities compatible with those observed in the Western Pacific, but we have also ran additional tests to verify our first-order conclusions are unchanged if we vary these values (Supplementary Figs. 1, 2).

In the single subduction model, the Pacific trench advances slowly to the left/west at ~ 2 cm/yr and far-field Eurasia is near-stationary (Fig. 3e, Supplementary Fig. 3). Pacific Plate motion is trench-perpendicular and at ~ 8 -10 cm/yr with respect to far-field Eurasia. These modeled velocities exhibit first-order agreement with observed plate velocities in seafloor-spreading aligned^{40,48} or fixed Pacific hotspot reference frames⁴³. Adding the Ryukyu-Nankai subduction zone induces a westward Pacific trench advance of 3-4 cm/yr adjacent to the Philippine Sea Plate (i.e., Izu-Bonin-Mariana) (Fig. 3f, Supplementary Figs. 1, 4). This is consistent with observed trench motions⁴⁸ and previous subduction modeling^{13,52,66} (Fig. 1a). These agreements suggest that our models capture the relevant large-scale geodynamic processes and are, in turn, suitable for investigating changes in the large-scale lithospheric stress state induced by double subduction.

Our numerical models show that the establishment of a double subduction system significantly influences the upper plate stress state in the Japan triple junction and immediately north of this region (Fig. 3e-g and Supplementary Figs. 4, 5). Without double subduction, the lithosphere in the Japan triple junction region is under relatively moderate trench-perpendicular compression of ~ 10 -20 MPa (Fig. 3e and Supplementary Fig. 4a). In such geodynamic models, this absolute upper plate stress state is largely controlled by the imposed upper plate size, thickness, density, and strength (e.g.,^{67,68}). Here, we therefore focus on the relative change in stress state induced by double subduction, as opposed to absolute stress values. Throughout the upper plate, double subduction induces a significant increase in the magnitude of trench-perpendicular

compressive stress (Fig. 3f and Supplementary Fig. 4b). Approximately 350 km onboard of the modeled Japan trench, i.e., in the vicinity of thrusting in Northeast Japan, the average trench-perpendicular compressive normal stress in the lithosphere doubles from ~ 20 MPa to ~ 40 MPa (Fig. 3g and Supplementary Fig. 5). The other normal stress components (vertical and trench-parallel) are only minimally affected by double subduction (Supplementary Fig. 4). This increase in trench-perpendicular compression is due to the along-strike transmission of the Philippine Sea Plate slab pull at the Pacific margin^{15,52}, which causes the Pacific slab to indent into the Eurasian upper plate at the Northeast Japan trench, thereby inducing enhanced upper plate compression. The region experiencing increases in horizontal compressive stress extends significantly, covering an area $\sim 1,000$ – $1,200$ km northward along the Pacific trench and an additional $\sim 1,300$ – $1,500$ km inland (Fig. 3g). This observation aligns with geological and geophysical evidence of compression, not only within the Japan arc but also in the backarc domain, which includes a deformed belt approximately $\sim 1,200$ km in length^{17–20,25–28,30–32} (Figs. 3g and 1a). The model results also highlight subtle compression extending north of northeast Japan, consistent with structural-geomorphological, seismological, and GPS data, all of which indicate mild compression/transpression along Sakhalin Island^{18,69} (Fig. 1a). The observation that the increased compressional lithospheric stress in the models does not extend beyond approximately $\sim 1,500$ km aligns with newly derived strain maps of eastern Eurasia, which indicate negligible active deformation west of the analyzed area⁷⁰ (Figs. 3g and 1a).

Additional tests also demonstrate that, irrespective of the chosen crustal and lithospheric strengths, an increase in trench-perpendicular horizontal compressive stress is a robust feature in transitioning from single to double subduction (Supplementary Figs. 1, 2). The magnitude of the increase

in horizontal compression varies with these properties – increasing for higher lithosphere viscosity or lower crustal viscosity – but the single-to-double stress increase is always prevalent. Hence, these results support the double subduction-induced orogeny hypothesis and provide an integrative geodynamic framework that links the evolution of the Ryukyu/Izu-Bonin-Marianas SDDS system^{2,5,13}, the motion of the Philippine Sea Plate, the associated westward drag of the Pacific trench^{45,49,50}, and the onset of arc-backarc compression in Northeast Japan^{17–20,25–28,32}. We suggest that this process accounts for the transition from soft to hard deformation during Izu arc subduction between 8 and 6 Myr ago²⁰.

Double subduction-induced orogeny in the geological record

Despite the substantial variability in scale and geometrical configuration inherent to highly dynamic double subduction systems, the documentation of active double subduction-induced orogeny in Northeast Japan has significant implications for the orogenic evolution of plate margins proximal to other ancient SDDS systems.

We propose double subduction-induced orogeny, generalized in Fig. 4a, offers a plausible explanation for the widespread Cretaceous compression documented west of the contemporaneous Neotethys subduction system^{71,72} (Fig. 4b). Here, a major change in absolute plate motion direction and the rapid northward drift of the African plate have been invoked to explain Late Cretaceous compression in Northern Africa⁷¹ (~86–84 Myr ago, Ayyubid Orogen⁷³), subduction initiation along the western Mediterranean region¹⁴, and far-field deformation in Central Europe (~84–75 Myr, Subhercinian Basin intraplate deformation⁷⁴) (Fig. 4b). Recent studies suggest that the onset of a

~12,000 km long SDDS at approximately 105 Myr⁷², along with the initiation of effective slab pull around ~95 Myr, coincided with the directional shift and acceleration in the drift of the African plate¹⁴. Updated plate kinematic data indicate that Africa–Eurasia relative plate motion along the western Mediterranean active margin transitioned from very slow motion (<5 mm/yr) in a south-southeast direction between ~125 and 90 Myr ago to a northeast direction with a velocity of 30–20 mm/yr from ~95 to 70 Myr ago¹⁴ (Fig. 4b). Although differing in dimensions, exact configuration, and partially in the nature of the orogenic processes it involves—such as the collision of Iberia—we propose that, within the framework of the double subduction-induced orogeny concept introduced here (Fig. 4a), a potential genetic connection exists (Fig. 4b). Specifically, this connection links the lifespan (105–75 Myr ago) of the large neighboring SDDS zone in the Neotethys Ocean with the associated plate kinematic change and the plate-wide compression (~90–75 Myr ago^{14,72}) observed to the west of this system (Fig. 4b).

Although more speculative due to the limited precision of early Paleozoic plate margin reconstructions for southwest Gondwana, another potential candidate for ancient double subduction-induced orogeny is the Paleozoic southern South American margin. Notably, the Chaitenia SDDS (~380–340 Myr ago) to the west of Patagonia⁷⁵ partially overlaps in time with the Chanic orogeny (~385–350 Myr ago) immediately to the north of this ancient SDDS⁷⁶. If active subduction characterized the entire margin at this time, as recently suggested⁷⁷, and there was a north-south mechanical connection between the Chaitenia subduction and that to the north, it is possible that the onset of the Chaitenia SDDS may have contributed to sustaining, if not driving, the protracted Chanic orogeny north of Patagonia. Although this hypothesis provides

an elegant explanation for the required trench advance to drive backarc closure of the Cuyano-*proto-ocean* basin, it warrants further investigation in light of more precise information on the nature and configuration of plate margins during the early Paleozoic.

The case of double subduction-induced orogeny in Japan illustrates how subduction interactions, linked to the emergence of a nearby subduction zone, can trigger far-field effects that cascade onto neighboring plate margins. This new perspective offers valuable insight into the interpretation of the seismotectonic setting of Northeast Japan, a region renowned for its devastating seismic events, including the Tohoku megathrust earthquake (2011, $M_w = 9.0$)³⁵ and the recent Noto Peninsula intraplate earthquake (2024, $M_w = 7.6$)³³ in the backarc area. With this understanding of the tectonic impact of SDDS zones, we anticipate that a more thorough examination of ancient plate margin geological records will uncover additional cases of double subduction zone-induced orogeny.

Methods

Numerical subduction modeling

Our 3-D subduction models were constructed using the ASPECT finite element code (version 2.5.0)^{53–56}. ASPECT was used to solve the conservation of momentum (Stokes equation) and mass (continuity equation) for an incompressible viscous fluid (Boussinesq approximation). Our models are purely compositional. That is, we neglect temperature/thermal diffusion and density and viscosity are purely dependent on compositional fields. Our models contain four compositional fields that have distinct material properties: the crust, upper plate (Eurasia) lithosphere, subducting plate (Philippine Sea Plate and Pacific Plate) lithosphere, and stiffer material placed at the lateral edges of the Pacific and Eurasian plates (e.g., see Supplementary Fig. 3a,c). The background material corresponds to the sub-lithospheric mantle and has another set of mechanical properties (see later paragraphs of this section). The lateral plate edges are imposed to be stiffer (higher viscosity) to retard subduction initiation in these locations. Our models are “pseudo-instantaneous” in that we solve for the velocity and stress fields associated with the imposed slab and plate tectonic geometries, as opposed to allowing the structure/slabs to advect for millions of years. In practice, we extract the solution after 10,000 years of model evolution. Given modeled plate velocities of ≤ 10 cm/yr, we do not observe noticeable advection of the mantle/tectonic plates over this duration.

We use ASPECT’s Adaptive mesh refinement (AMR) functionality to highly refine the finite element mesh within the lithospheric plates – including the crustal layer that makes up the plate boundary shear zone – and within other regions containing large viscosity gradients (e.g., the upper-to-lower mantle transition). In these regions, we have a maximum resolution corresponding to ≈ 4 km wide cubic finite elements. The lowest resolution occurs far away

from the tectonic plates and corresponds to ≈ 60 km wide cubic finite elements. This approach results in $\sim 5,500,000$ cells unequally distributed throughout the model domains. Simulations were ran on the Skylake (SKX) nodes of the Texas Advanced Computing Center's Stampede3 cluster.

Our subduction models have a Cartesian domain spanning 8000 km (trench-perpendicular) $\times$ 5000 km (trench-parallel) $\times$ 1000 km (depth). There are no external forces or velocities applied to the subduction system and all model boundaries are mechanically free slip. In our single subduction case, we model Pacific subduction beneath a homogeneous Eurasian plate (Fig. 3a). In the double subduction case, we add the Philippine Sea plate and Ryukyu-Nankai subduction, but leave the Pacific plate/slab geometry completely unchanged (Fig. 3b, Supplementary Fig. 6). At the northern edge of the subduction system ($y = 5000$ km), we impose a “free” plate boundary (i.e., tectonic plates are not attached to the model boundary; Supplementary Fig. 3a,c). This enables the Pacific plate to have a north-south velocity component in this northerly model region. At the other side of the subduction system ($y = 0$), the plates are “fixed” to the model boundary. This boundary acts as a plane of symmetry, thereby enabling us to model the diamond-shaped Philippine Sea plate as a triangle (Fig. 3a,b and Supplementary Fig. 3).

We constructed the input plate and slab geometries using the Geodynamic World Builder tool^{78,79}. We included only the tectonic complexities that are of first-order importance for our proposed mechanism. Our plate boundaries are linear, aside from the protruding cusp of the Pacific trench in northern Japan. This north south segment is 1000 km long and at an angle of 27° relative to the 1600 km-long northern (Kuril-Kamchatka) and southern (Izu-Bonin) Pacific sections (Fig. 1a). In the double subduction case, the Philippine Sea plate was added by including the Ryukyu-Nankai trenches (Fig. 1a). This Philippine Sea

Plate-Eurasia plate boundary is 1700 km long and at 29° to the Japan trench. We adopt a constant Pacific slab geometry from north-to-south, with the base of the slab extending to 625 km depth. Of the 1700 km long Philippine Sea Plate-Eurasia plate boundary – in the double subduction system – the southern 1250 km corresponds to deeper Ryukyu subduction (400 km slab depth) and the northern 450 km corresponds to shallower Nankai subduction (210 km slab depth)^{44,57,58}. Detailed slab geometries are displayed in Supplementary Figure 3 (cross-sections) and Figure 6 (3-D visualization). Both the Pacific and Philippine Sea Plate slabs have a slab-top radii of curvature of 185 km and, below this curved portion, dips of 60 deg. To approximate the interaction of the Pacific slabs with the higher viscosity lower mantle, we linearly reduce the dip to zero as a function of along-slab distance along the deepest 200 km of the slab. We impose lithospheric thicknesses of 90 km and 60 km for the older and younger Pacific and Philippine Sea Plates, respectively, and a 60 km thickness for the idealized Eurasian plate. A weak, 10 km thick crustal layer is placed on top of all plates and along the upper surface of slabs to 70 km depth. This facilitates plate boundary decoupling, thereby enabling us to reproduce the main features of observed plate and trench velocities (e.g., Fig. 1a,^{40,43,48}). To ensure the Philippine Sea Plate (and Nankai slab) and Eurasian plate are also decoupled, a ≈ 150 km long and 50 km wide weak zone, with the same properties as the background mantle, is placed at the northern edge of the Philippine Sea Plate and Nankai slab. This weak zone extends through the entire plate and has two components: an east-west (x-directed) component that delineates the northern/top edge of the Philippine Sea Plate, and a southeast-northwest component that is parallel to the lateral edge of the northwest dipping Nankai slab (see Supplementary Fig. 6 for a 3-D view of the weak zone). The latter is needed to stop the Nankai slab from intersecting/coupling with the overriding

Eurasia. To check that the specific geometry of this weak zone does not impact first order model behavior, we ran an additional double subduction model with a wider weak region in the triple junction region. In this case, weak material fills the entire region (triangle) between the northern edge of the Nankai slab and its projection onto the Japan Trench. Neither the stress pattern or stress magnitudes are significantly affected (cf. Supplementary Figs. 4b,c).

The Pacific and Philippine Sea lithospheric plates have constant densities of 3350 kg/m^3 ; 50 kg/m^3 denser than the sub-lithospheric mantle. The Eurasian plate and crustal layers are neutrally buoyant with respect the mantle, with densities of 3300 kg/m^3 . This is to approximate the lower densities of continental lithosphere and basaltic crust relative to oceanic lithosphere. The lithospheric and crustal layers are isoviscous and the underlying sub-lithospheric mantle has a composite Newtonian and power-law rheology (e.g.,⁸⁰) to mimic the first-order effects of diffusion and dislocation creep, respectively. Lithospheric plates are set to be 500 times more viscous ($5 \times 10^{22} \text{ Pas}$) than the reference mantle viscosity (10^{20} Pas). This is within the 10-1000 range of slab-mantle viscosity contrasts generally suggested by geoid constraints^{61,62}, slab shapes and trench motions^{60,81,82}, and plate bending considerations^{63,83,84}. We consider isoviscous plates, i.e., without plastic yielding, to both simplify our model setups and minimize non-linearity (and hence speed up model convergence). While this limits lithospheric deformation to diffuse viscous deformation distributed over 100s-to-1000s of km, as opposed to more "Earth-like" localized shear zones, it enables us to examine which regions experience high viscous stress and hence would ultimately be prone to yielding. We increase the lithosphere viscosity by an order of magnitude within 1000 km wide (trench-perpendicular) edges of the Eurasian and Pacific Plates; this is to ensure subduction does not initiate at the edges. This higher plate

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edge viscosity causes the stress discontinuities noticeable at $x = 1000$ km and 6000 km (Fig. 3e,f). The crustal layer has a viscosity of 5×10^{19} Pas. Considering a deep plate interface temperature of, e.g., 500-600°C, and strain rate of a 10^{-12} s^{-1} , this viscosity overlaps with that recorded for blueschists, and resides between that of stronger meta-basalts and weaker metasediments and serpentinite^{64,85–88}. It is also within the 10^{18} - 10^{20} Pas range estimated using numerical shear experiments for the global range of exhumed shear zones lithological distributions⁶⁵. However, given our simplified shear zone implementation (a single isoviscous layer), the most appropriate effective shear zone viscosity is uncertain, we therefore test a range of values (Supplementary Fig. 1).

Power-law flow in the sub-lithospheric mantle is included due to its large effect on plate and trench motions (e.g.,^{89,90}). Due to the compositional nature of our models, we neglect the temperature dependence of mantle viscosity, and additionally omit the pressure dependence. The Newtonian viscosity component of the mantle, η_N , is set to 2×10^{20} Pas. The power-law component, η_P , is calculated following:

$$\eta_P = \eta_N (\dot{\epsilon} / \dot{\epsilon}_T)^{\frac{1-n}{n}} \quad (1)$$

where $\dot{\epsilon}$ is the square root of the second invariant of the strain rate tensor, n is the power-law exponent ($= 3.5$), and $\dot{\epsilon}_T$ is the transition strain rate. Following Holt and Becker⁹⁰, we set $\dot{\epsilon}_T = 5 \times 10^{-15} \text{ s}^{-1}$. For strain rates at this value, $\eta_P = \eta_N = 2 \times 10^{20}$ Pas. The effective model viscosity is then calculated as:

$$\eta_{eff} = \left(\frac{1}{\eta_N} + \frac{1}{\eta_P} \right)^{-1} \quad (2)$$

which produces the reference mantle viscosity of $\eta_{eff} = 10^{20}$ Pas at $\dot{\epsilon} = 5 \times 10^{-15} \text{ s}^{-1}$. Given that significant uncertainty remains regarding the

appropriate creep law parameters (e.g., ⁹¹), even for the dominant upper mantle mineral olivine, we consider our simplified mantle rheology to be appropriately complex. Beyond 660 km depth, the lower mantle is set to a viscosity of 5×10^{21} Pas (i.e., a factor 50 increase relative to the reference upper mantle viscosity). This follows geoid and slab sinking rate studies that indicate the lower mantle has a higher viscosity than the upper mantle (e.g., ^{92,93}).

To verify the first-order behavior of our models does not depend on the assigned crustal and lithospheric viscosities, we have ran additional models with different values. Relative to the reference crustal viscosity (5×10^{19} Pas), we test a lower value of 10^{19} Pas and higher values of 2.5×10^{20} Pas and 10^{20} Pas (Supplementary Fig. 2). Relative to the reference slab-mantle viscosity ratio (500), we test higher and lower values of 2000 and 125 (Supplementary Fig. 3). For each of these additional parameter tests, we turn off power-law flow (i.e., Equation 1) within the sub-lithospheric mantle and instead use a constant (Newtonian) mantle viscosity of 10^{20} Pas Pas. Removing this complexity does not substantially affect the modeled stress fields (cf. Fig. 3e-g and Supplementary Figs. 2 and 3), but reduces model run times to $\approx 30\%$ of the equivalent power-law models. Overall, increasing the slab-mantle viscosity ratio and/or decreasing the crustal viscosity increase the magnitude of the double subduction induced stress increase. However, there is always an increase in horizontal compression due to double subduction.

Declarations

Data availability

All data needed to evaluate the conclusions in the paper are presented in this manuscript, the Supplementary information, and the online repository containing the model input files.

Code availability

The ASPECT code used for our numerical modeling is freely available at <https://aspect.geodynamics.org/>. All files needed to reproduce the models are contained in the following GitHub repository: <https://github.com/AdamFHolt/Japan-Compression>. Upon (potential) paper acceptance, these files will be transferred to a permanent Zenodo repository.

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Author contributions

G. M. G. conceived the study. Z. G. and A. F. H. designed and carried out the numerical modeling. C. F. provided background on the study area and geodynamic processes. All authors contributed to writing and editing the manuscript, as well as creating the figures.

References

- [1] Stern, R. J. Subduction zones. *Rev. Geophys.* **40** (4), 3–1 (2002) .
- [2] Holt, A., Royden, L. & Becker, T. The dynamics of double slab subduction. *Geophys. J. Int.* **209** (1), 250–265 (2017) .

- [3] Pusok, A. E. & Stegman, D. R. Formation and stability of same-dip double subduction systems. *J. Geophys. Res. Solid Earth* **124** (7), 7387–7412 (2019) .
- [4] Király, Á., Holt, A. F., Funiciello, F., Faccenna, C. & Capitanio, F. A. Modeling slab-slab interactions: Dynamics of outward dipping double-sided subduction systems. *Geochem. Geophys. Geosyst.* **19** (3), 693–714 (2018) .
- [5] Faccenna, C., Holt, A. F., Becker, T. W., Lallemand, S. & Royden, L. H. Dynamics of the Ryukyu/Izu-Bonin-Marianas double subduction system. *Tectonophysics* **746**, 229–238 (2018) .
- [6] Jagoutz, O., Royden, L., Holt, A. F. & Becker, T. W. Anomalously fast convergence of India and Eurasia caused by double subduction. *Nat. Geosci.* **8** (6), 475–478 (2015) .
- [7] Mishin, Y. A., Gerya, T. V., Burg, J.-P. & Connolly, J. A. Dynamics of double subduction: Numerical modeling. *Phys. Earth Planet. Inter.* **171** (1-4), 280–295 (2008) .
- [8] Zhang, K., Liao, J. & Gerya, T. Onset of double subduction controls plate motion reorganisation. *Nat. Comm.* **15** (1), 1–9 (2024) .
- [9] Zhang, Q., Guo, F., Zhao, L. & Wu, Y. Geodynamics of divergent double subduction: 3-D numerical modeling of a Cenozoic example in the Molucca Sea region, indonesia. *J. Geophys. Res. Solid Earth* **122** (5), 3977–3998 (2017) .

- [10] Soesoo, A., Bons, P. D., Gray, D. R. & Foster, D. A. Divergent double subduction: tectonic and petrologic consequences. *Geology* **25** (8), 755–758 (1997) .
- [11] Pindell, J. L. *et al.* A plate-kinematic framework for models of Caribbean evolution. *Tectonophysics* **155** (1-4), 121–138 (1988) .
- [12] Santosh, M. A synopsis of recent conceptual models on supercontinent tectonics in relation to mantle dynamics, life evolution and surface environment. *J. Geodyn.* **50** (3-4), 116–133 (2010) .
- [13] Čížková, H. & Bina, C. R. Geodynamics of trench advance: Insights from a Philippine-Sea-style geometry. *Earth Planet. Sci. Lett.* **430**, 408–415 (2015) .
- [14] Gürer, D., Granot, R. & van Hinsbergen, D. J. Plate tectonic chain reaction revealed by noise in the Cretaceous quiet zone. *Nat. Geosci.* **15** (3), 233–239 (2022) .
- [15] Jian, H. *et al.* Slab pull drives IBM Trench advance despite the weakened Philippine Sea Plate. *Geophys. Res. Lett.* **50** (21), e2023GL106554 (2023) .
- [16] Dewey, J. F. in *Episodicity, sequency and style at convergent plate boundaries* (ed.Strangway, D. W.) *The Continental Crust and its Mineral Deposits*, Vol. 20 of *Special Paper* 553–573 (Geological Association of Canada, 1980).
- [17] Sato, H. The relationship between late Cenozoic tectonic events and stress field and basin development in northeast Japan. *J. Geophys. Res. Solid Earth* **99** (B11), 22261–22274 (1994) .

- [18] Jolivet, L., Tamaki, K. & Fournier, M. Japan Sea, opening history and mechanism: A synthesis. *J. Geophys. Res. Solid Earth* **99** (B11), 22237–22259 (1994) .
- [19] Lallemand, S. & Jolivet, L. Japan Sea: a pull-apart basin? *Earth Planet. Sci. Lett.* **76** (3-4), 375–389 (1986) .
- [20] Taira, A. Tectonic evolution of the Japanese island arc system. *Ann Rev Earth Planet Sci* **29** (1), 109–134 (2001) .
- [21] Kano, K., Uto, K. & Ohguchi, T. Stratigraphic review of Eocene to Oligocene successions along the eastern Japan Sea: Implication for early opening of the Japan Sea. *J. Asian Earth Sci.* **30** (1), 20–32 (2007) .
- [22] Kimura, G. & Tamaki, K. Collision, rotation, and back-arc spreading in the region of the Okhotsk and Japan Seas. *Tectonics* **5** (3), 389–401 (1986) .
- [23] Tamaki, K. Geological structure of the Japan Sea and its tectonic implications. *Chishitsu Chosasho Geppo;(Japan)* **39** (5) (1988) .
- [24] Ingle, J. J. Subsidence of the Japan Sea: stratigraphic evidence from odp sites and onshore sections. *Proc. Ocean Drill. Program Sci. Results* (1992) .
- [25] Okada, S. & Ikeda, Y. Quantifying crustal extension and shortening in the back-arc region of Northeast Japan. *J. Geophys. Res. Solid Earth* **117** (B1) (2012) .
- [26] Regalla, C., Fisher, D. & Kirby, E. Timing and magnitude of shortening within the inner fore arc of the Japan Trench. *J. Geophys. Res. Solid*

- Earth* **115** (B3) (2010) .
- [27] Tamaki, K. & Honza, E. Incipient subduction and deduction along the eastern margin of the Japan Sea. *Tectonophysics* **119** (1-4), 381–406 (1985) .
- [28] Kim, G.-B., Yoon, S.-H., Kim, S.-S. & So, B.-D. Transition from buckling to subduction on strike-slip continental margins: Evidence from the East Sea (Japan Sea). *Geology* **46** (7), 603–606 (2018) .
- [29] Iwasaki, T. *et al.* Extensional structure in northern Honshu Arc as inferred from seismic refraction/wide-angle reflection profiling. *Geophys. Res. Lett.* **28** (12), 2329–2332 (2001) .
- [30] Wei, D. & Seno, T. in *Determination of the amurian plate motion* (eds Flower, M. F. J., Chung, S.-L., Lo, C.-H. & Lee, T.-Y.) *Mantle Dynamics and Plate Interactions in East Asia*, Vol. 27 of *Geodynamic Series* 337–346 (American Geophysical Union (AGU), 1998).
- [31] Naramura, K. Possible Nascent Trench along the Eastern Japan Sea as the Convergent Boundary between Eurasian and North American Plates. *Bull. Earthq. Res. Inst., Univ. Tokyo* **58**, 711–722 (1983). URL <https://doi.org/10.15083/0000032948>. <https://doi.org/10.15083/0000032948> .
- [32] van der Werff, W. Backarc deformation along the eastern Japan Sea margin, offshore northern Honshu. *J. Asian Earth Sci.* **18** (1), 71–95 (2000) .
- [33] Fujii, Y. & Satake, K. Slip distribution of the 2024 Noto Peninsula earthquake (MJMA 7.6) estimated from tsunami waveforms and GNSS data. *Earth Planets Space* **76** (1), 44 (2024) .

- [34] Mazzotti, S., Henry, P. & Le Pichon, X. Transient and permanent deformation of central Japan estimated by GPS: 2. strain partitioning and arc–arc collision. *Earth Planet. Sci. Lett.* **184** (2), 455–469 (2001) .
- [35] Ikeda, Y. Strain buildup in the Northeast Japan orogen with implications for gigantic subduction earthquakes. *Episodes J. Int. Geosci.* **37** (4), 234–245 (2014) .
- [36] Pollitz, F. F. Pliocene change in Pacific-plate motion. *Nature* **320** (6064), 738–741 (1986) .
- [37] Lallemand, S., Heuret, A. & Boutelier, D. On the relationships between slab dip, back-arc stress, upper plate absolute motion, and crustal nature in subduction zones. *Geochem. Geophys. Geosyst.* **6** (9) (2005) .
- [38] Schellart, W. P. Overriding plate shortening and extension above subduction zones: A parametric study to explain formation of the Andes mountains. *Geol. Soc. Am. Bull.* **120** (11-12), 1441–1454 (2008) .
- [39] Müller, R. D. *et al.* A global plate model including lithospheric deformation along major rifts and orogens since the Triassic. *Tectonics* **38** (6), 1884–1907 (2019) .
- [40] Becker, T. W., Schaeffer, A. J., Lebedev, S. & Conrad, C. P. Toward a generalized plate motion reference frame. *Geophys. Res. Lett.* (2015). <https://doi.org/10.1002/2015GL063695> .
- [41] O'Neill, C., Müller, D. & Steinberger, B. On the uncertainties in hot spot reconstructions and the significance of moving hot spot reference frames. *Geochem. Geophys. Geosyst.* **6** (4) (2005) .

- [42] DeMets, C., Gordon, R. G., Argus, D. F. & Stein, S. Effect of recent revisions to the geomagnetic reversal time scale on estimates of current plate motions. *Geophys. Res. Lett.* **21** (20), 2191–2194 (1994) .
- [43] Gripp, A. E. & Gordon, R. G. Young tracks of hotspots and current plate velocities. *Geophys. J. Int.* **150** (2), 321–361 (2002) .
- [44] Wu, J., Suppe, J., Lu, R. & Kanda, R. Philippine Sea and East Asian plate tectonics since 52 Ma constrained by new subducted slab reconstruction methods. *J. Geophys. Res. Solid Earth* **121** (6), 4670–4741 (2016) .
- [45] Takahashi, M. The cause of the east–west contraction of Northeast Japan. *J. Geol. Soc. Jpn.* **68** (4), 155–161 (2017) .
- [46] Deschamps, A. & Lallemand, S. The West Philippine Basin: An Eocene to early Oligocene back arc basin opened between two opposed subduction zones. *J. Geophys. Res. Solid Earth* **107** (B12), EPM–1 (2002) .
- [47] Ficini, E., Cuffaro, M., Doglioni, C. & Gerya, T. Variable plate kinematics promotes changes in back-arc deformation regime along the north-eastern Eurasia plate boundary. *Sci. Rep.* **14** (1), 7220 (2024) .
- [48] Heuret, A. & Lallemand, S. Plate motions, slab dynamics and back-arc deformation. *Phys. Earth Plan. Int.* (2005). <https://doi.org/10.1016/j.pepi.2004.08.022> .
- [49] Carlson, R. & Mortera-Gutiérrez, C. Subduction hinge migration along the Izu-Bonin-Mariana arc. *Tectonophysics* **181** (1-4), 331–344 (1990) .
- [50] Faccenna, C., Di Giuseppe, E., Funiciello, F., Lallemand, S. & van Hunen, J. Control of seafloor aging on the migration of the Izu–Bonin–Mariana

- trench. *Earth Planet. Sci. Lett.* **288** (3–4), 386–398 (2009) . 1289
1290
- [51] Moreno, E. *et al.* Numerical modeling of subduction and evaluation of 1291
Philippine Sea Plate tectonic history along the Nankai Trough. *Sci. Rep.* 1292
13 (1), 18313 (2023) . 1293
1294
1295
1296
- [52] Holt, A. F., Royden, L. H., Becker, T. W. & Faccenna, C. Slab interactions 1297
in 3-D subduction settings: The Philippine Sea Plate region. *Earth Planet.* 1298
Sci. Lett. **489**, 72–83 (2018) . 1299
1300
1301
1302
- [53] Bangerth, W. *et al.* ASPECT v2.5.0 (2023). URL <https://doi.org/10.5281/zenodo.8200213>. 1303
1304
1305
1306
- [54] Bangerth, W. *et al.* ASPECT: Advanced Solver for Problems in Earth’s 1307
ConvecTion, User Manual (2023). URL [https://doi.org/10.6084/m9.](https://doi.org/10.6084/m9.figshare.4865333) 1308
[figshare.4865333](https://doi.org/10.6084/m9.figshare.4865333). Doi:10.6084/m9.figshare.4865333. 1309
1310
1311
1312
- [55] Kronbichler, M., Heister, T. & Bangerth, W. High accuracy mantle con- 1313
vection simulation through modern numerical methods. *Geophys. J. Int.* 1314
191 (1), 12–29 (2012) . 1315
1316
1317
1318
- [56] Heister, T., Dannberg, J., Gassmöller, R. & Bangerth, W. High accu- 1319
racy mantle convection simulation through modern numerical methods–II: 1320
realistic models and problems. *Geophys. J. Int.* **210** (2), 833–851 (2017) . 1321
1322
1323
1324
- [57] Hayes, G. P. *et al.* Zone Geometry Model. *Science* **362** (6410), 58–61 1325
(2018) . 1326
1327
1328
- [58] Zhao, D., Wang, J., Huang, Z. & Liu, X. Seismic structure and subduction 1329
dynamics of the western Japan arc. *Tectonophysics* **802** (2021). [https:](https://doi.org/10.1016/j.tecto.2021.228743) 1330
[//doi.org/10.1016/j.tecto.2021.228743](https://doi.org/10.1016/j.tecto.2021.228743) . 1331
1332
1333
1334

- [59] Auer, L., Boschi, L., Becker, T., Nissen-Meyer, T. & Giardini, D. Savani: A variable resolution whole-mantle model of anisotropic shear velocity variations based on multiple data sets. *J. Geophys. Res. Solid Earth* **119** (4), 3006–3034 (2014) .
- [60] Funiciello, F. *et al.* Trench migration, net rotation and slab–mantle coupling. *Earth Planet. Sci. Lett.* **271** (1-4), 233–240 (2008). <https://doi.org/10.1016/j.epsl.2008.04.006> .
- [61] Moresi, L. & Gurnis, M. Constraints on the lateral strength of slabs from three-dimensional dynamic flow models. *Earth Planet. Sci. Lett.* **138** (1), 15–28 (1996). [https://doi.org/https://doi.org/10.1016/0012-821X\(95\)00221-W](https://doi.org/https://doi.org/10.1016/0012-821X(95)00221-W) .
- [62] Mao, W. & Zhong, S. Constraints on Mantle Viscosity From Intermediate-Wavelength Geoid Anomalies in Mantle Convection Models With Plate Motion History. *J. Geophys. Res. Solid Earth* **126** (4), e2020JB021561 (2021). <https://doi.org/https://doi.org/10.1029/2020JB021561> .
- [63] Wu, B., Conrad, C. P., Heuret, A., Lithgow-Bertelloni, C. & Lallemand, S. Reconciling strong slab pull and weak plate bending: The plate motion constraint on the strength of mantle slabs. *Earth Planet. Sci. Lett.* **272** (1-2), 412–421 (2008). <https://doi.org/10.1016/j.epsl.2008.05.009> .
- [64] Behr, W. M. & Becker, T. W. Sediment control on subduction plate speeds. *Earth Planet. Sci. Lett.* **502**, 166–173 (2018). <https://doi.org/10.1016/j.epsl.2018.08.057> .
- [65] Abila, A. L., Behr, W. M. & Ruh, J. Strength of viscous subduction interfaces: A global compilation. *Geology* (2024). <https://doi.org/https://doi.org/10.1130/G51117.1> .

- [//doi.org/10.1130/G52518.1](https://doi.org/10.1130/G52518.1) . 1381
1382
- [66] Faccenna, C., Oncken, O., Holt, A. F. & Becker, T. W. Initiation of the 1383
Andean orogeny by lower mantle subduction. *Earth Planet. Sci. Lett.* 1384
463, 189–201 (2017) . 1385
1386
1387
1388
- [67] Capitanio, F. A., Stegman, D. R., Moresi, L. N. & Sharples, W. Upper 1389
plate controls on deep subduction, trench migrations and deformations 1390
at convergent margins. *Tectonophysics* (2010). [https://doi.org/10.1016/](https://doi.org/10.1016/j.tecto.2009.08.020) 1391
[j.tecto.2009.08.020](https://doi.org/10.1016/j.tecto.2009.08.020) . 1392
1393
1394
1395
1396
- [68] Holt, A., Becker, T. & Buffett, B. Trench migration and overriding plate 1397
stress in dynamic subduction models. *Geophys. J. Int.* **201** (1) (2015). 1398
<https://doi.org/10.1093/gji/ggv011> . 1399
1400
1401
1402
- [69] Bogomolov, L., Sim, L. & Kamenev, P. in *Neotectonics and stressed state* 1403
patterns of the Sakhalin Island (IntechOpen, 2020). 1404
1405
1406
- [70] Hao, M., Li, Y. & Zhuang, W. Crustal movement and strain distribution 1407
in East Asia revealed by GPS observations. *Sci. Rep.* **9** (1), 16797 (2019) . 1408
1409
1410
- [71] Jolivet, L. *et al.* Neo-tethys geodynamics and mantle convection: from 1411
extension to compression in Africa and a conceptual model for obduction. 1412
Can. J. Earth Sci. **53** (11), 1190–1204 (2016) . 1413
1414
1415
1416
- [72] van Hinsbergen, D. J. *et al.* A record of plume-induced plate rotation 1417
triggering subduction initiation. *Nat. Geosc.* **14** (8), 626–630 (2021) . 1418
1419
1420
- [73] Şengör, A. C. & Stock, J. The Ayyubid Orogen: An ophiolite obduction- 1421
driven orogen in the late Cretaceous of the Neo-Tethyan south margin. 1422
Geosc. Can. **41** (2), 225–254 (2014) . 1423
1424
1425
1426

- [74] Kley, J. & Voigt, T. Late Cretaceous intraplate thrusting in central Europe: Effect of Africa-Iberia-Europe convergence, not Alpine collision. *Geology* **36** (11), 839–842 (2008) .
- [75] Rapela, C. W. *et al.* The Devonian accretionary orogen of the North Patagonian cordillera. *Gondwana Res.* **96**, 1–21 (2021) .
- [76] Willner, A. P. *et al.* The geodynamics of collision of a microplate (Chilenia) in Devonian times deduced by the pressure–temperature–time evolution within part of a collisional belt (Guarguaraz Complex, W-Argentina). *Contrib. Mineral. Petrol.* **162**, 303–327 (2011) .
- [77] Dahlquist, J. A. *et al.* A review of Devonian–Carboniferous magmatism in the central region of Argentina, pre-Andean margin of SW Gondwana. *Earth-Sci. Rev.* **221**, 103781 (2021) .
- [78] Fraters, M., Thieulot, C., van den Berg, A. & Spakman, W. The Geodynamic World Builder: a solution for complex initial conditions in numerical modeling. *Solid Earth* **10** (5), 1785–1807 (2019). URL <https://www.solid-earth.net/10/1785/2019/>. <https://doi.org/10.5194/se-10-1785-2019> .
- [79] Fraters, M. The Geodynamic World Builder (2020). URL <https://doi.org/10.5281/zenodo.3900603>.
- [80] Zhong, S., Gurnis, M. & Moresi, L. Role of faults, nonlinear rheology, and viscosity structure in generating plates from instantaneous mantle flow models. *J. Geophys. Res. Solid Earth* **103** (B7), 15255–15268 (1998). <https://doi.org/https://doi.org/10.1029/98JB00605> .

- [81] Stegman, D. R., Freeman, J., Schellart, W. P., Moresi, L. & May, D. Influence of trench width on subduction hinge retreat rates in 3-D models of slab rollback. *Geochem. Geophys. Geosyst.* (2006). <https://doi.org/10.1029/2005GC001056> .
- [82] Ribe, N. M. Bending mechanics and mode selection in free subduction: A thin-sheet analysis. *Geophys. J. Int.* (2010). <https://doi.org/10.1111/j.1365-246X.2009.04460.x> .
- [83] Conrad, C. P. & Hager, B. H. Effects of plate bending and fault strength at subduction zones on plate dynamics. *J. Geophys. Res. Solid Earth* (1999). <https://doi.org/10.1029/1999jb900149> .
- [84] Becker, T. W., Faccenna, C., O'Connell, R. J. & Giardini, D. The development of slabs in the upper mantle: Insights from numerical and laboratory experiments. *Journal of Geophysical Research: Solid Earth* **104** (B7), 15207–15226 (1999). <https://doi.org/10.1029/1999JB900140> .
- [85] Zhang, J. & Green, H. W. Experimental Investigation of Eclogite Rheology and Its Fabrics at high Temperature and Pressure. *J. Met. Geol.* **25** (2), 97–115 (2007). <https://doi.org/10.1111/j.1525-1314.2006.00684.x> .
- [86] Hilaiet, N. *et al.* High-pressure creep of serpentine, interseismic deformation, and initiation of subduction. *Science* **318**, 1910–1913 (2007) .
- [87] Hirth, G., Teyssier, C. & Dunlap, J. W. An evaluation of quartzite flow laws based on comparisons between experimentally and naturally

- deformed rocks. *International Journal of Earth Sciences* **90**, 77–87 (2001)
- .
- [88] Tokle, L., Hufford, L. J., Behr, W. M., Morales, L. F. G. & Madonna, C. Diffusion creep of sodic amphibole-bearing blueschist limited by microboudinage. *Journal of Geophysical Research: Solid Earth* **128** (9), e2023JB026848 (2023). <https://doi.org/https://doi.org/10.1029/2023JB026848> .
- [89] Čížková, H. & Bina, C. R. Effects of mantle and subduction-interface rheologies on slab stagnation and trench rollback. *Earth Planet. Sci. Lett.* (2013). <https://doi.org/10.1016/j.epsl.2013.08.011> .
- [90] Holt, A. & Becker, T. The effect of a power-law mantle viscosity on trench retreat rate. *Geophys. J. Int.* **208** (1) (2017). <https://doi.org/10.1093/gji/ggw392> .
- [91] King, S. D. Reconciling laboratory and observational models of mantle rheology in geodynamic modelling. *J. Geodyn.* (2016). <https://doi.org/10.1016/j.jog.2016.03.005> .
- [92] Hager, B. H. Subducted slabs and the geoid: constraints on mantle rheology and flow. *J. Geophys. Res. Solid Earth* **89**, 6003–6015 (1984) .
- [93] Ricard, Y., Richards, M. A., Lithgow-Bertelloni, C. & Le Stunff, Y. A geodynamic model of mantle density heterogeneity. *J. Geophys. Res. Solid Earth* **98**, 21895–21909 (1993) .

Additional information

Supplementary Information is available for this paper

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Figures

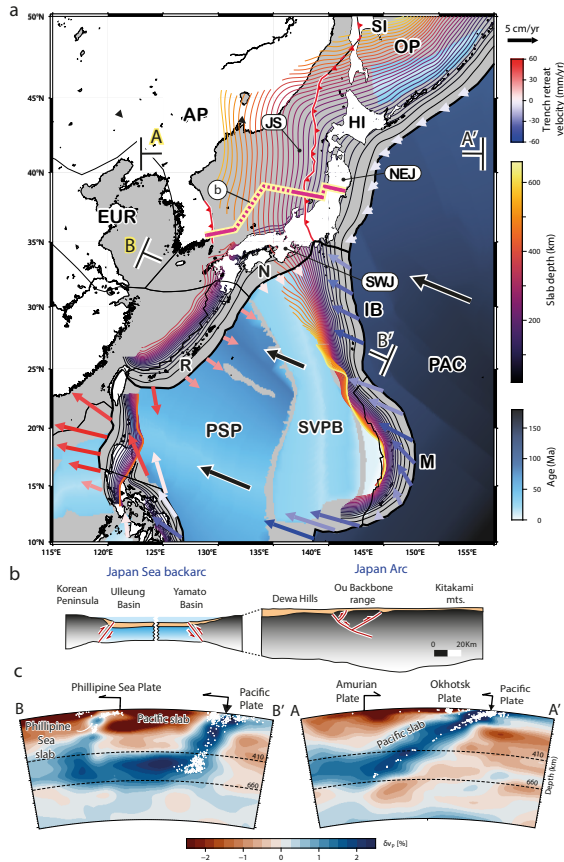


Fig. 1 Present-day plate kinematic setting and mantle structure of the western Pacific subduction zone. **a** Map indicating the major tectonic plates in the study area, subduction isodepths⁵⁷, absolute trench and upper plate kinematics⁴⁰, the location of Northeast Japan Island and the Japan Sea, and the Ryukyu-Nankai/Izu-Bonin-Marianas SDDS system. This map also shows the location of a composite structural cross-section from the northeast Japan arc²⁹ to the backarc region up to the eastern Korean margin²⁸ shown in **b**. **c** Seismic tomography sections⁵⁹ showing the mantle structure and slab seismicity associated with the Northeastern Japan subduction and the Ryukyu/Izu-Bonin-Marianas SDDS system. Abbreviations are: EUR: Eurasian Plate; AP: Amurian Plate; PSP: Philippine Sea Plate; PAC: Pacific Plate; OP: Okhotsk Plate; HI: Hokkaido Island; SI: Sakhalin Island; NEJ: Northeastern Japan; SWJ: Southwest Japan; JS: Japan Sea; R: Ryukyu trench; N: Nankai trench; IB: Izu-Bonin subduction segment; M: Marianas subduction segment; and SVPB: Extinct Shikoku-Parece-Vela backarc basin.

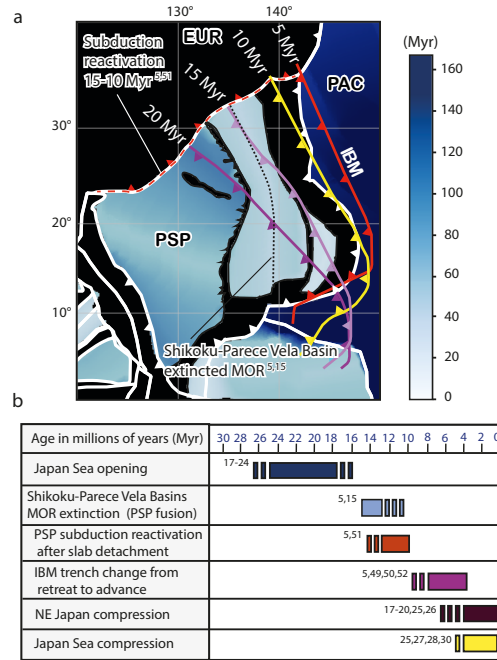


Fig. 2 Cenozoic kinematics of Izu-Bonin-Marianas trench and its relationship to deformation events in the arc and backarc regions of Northeast Japan. **a** Evolution of the Izu-Bonin-Marianas trench migration in the late Cenozoic from Faccenna *et al.*⁵ that considers a moving Indian/Atlantic hotspot reference frame⁴¹. **b** Time correlation of tectonic and kinematic changes in the study area. Please note that the progressive shift in the motion of the Izu-Bonin-Marianas Trench, following PSP fusion and the reactivation of subduction along the Ryukyu-Nankai Trench, slightly precedes and overlaps with the onset of strong compressional conditions in Northeast Japan and the Japan Sea back-arc basin. Abbreviations are: IBM: Izu-Bonin-Marianas Trench, EUR: Eurasian Plate; PSP: Philippine Sea Plate, and PAC: Pacific Plate.

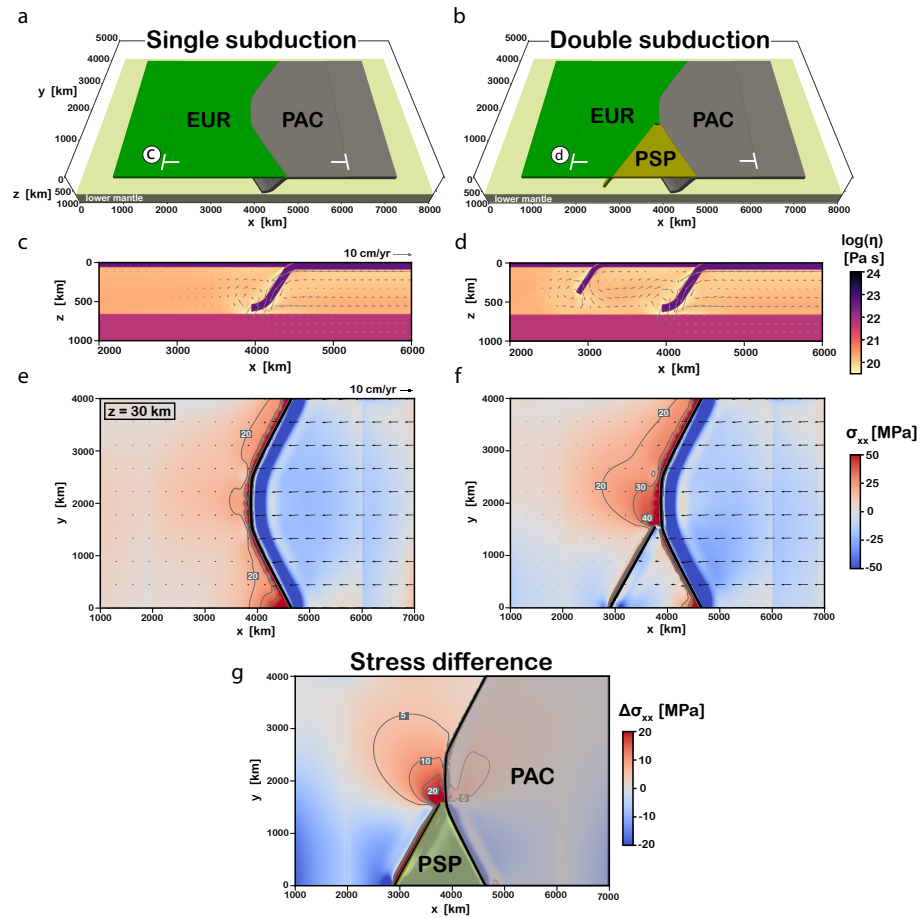


Fig. 3 3-D numerical subduction models testing the influence of the Ryukyu/Izu-Bonin-Marianas SDDS on driving upper plate compression in the northeastern arc-to-backarc regions of Japan. Set up for **a** single and **b** double subduction experiments, with the viscosity and velocity fields in cross sections shown in **c** and **d**, and the computed trench-perpendicular (x-direction) normal stress field and plate velocity displayed in **e** and **f** for the reference single and double subduction models, respectively. Stress and velocity is extracted within the lithospheric plates at 30 km depth. **g** Shows the normal stress difference between the models indicating significantly enhanced compression in the area corresponding to the location of Northeast Japan. For simplicity, the Okhotsk Plate is not included, and only an equivalent to the Eurasian Plate is considered. Abbreviations are: EUR: Eurasian Plate; PSP: Philippine Sea Plate, and PAC: Pacific Plate.

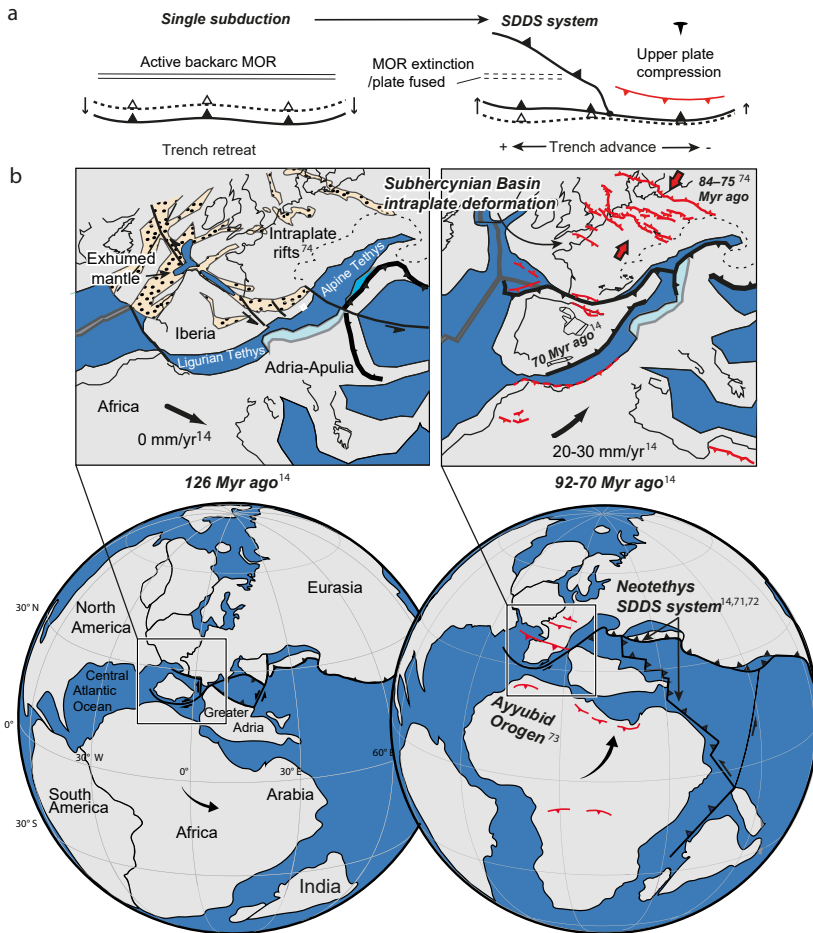


Fig. 4 Generalized conceptual model of double subduction-induced orogeny and a proposed Mesozoic case associated with the Neotethys SDSS system. **a** A conceptual model illustrates the impact on trench dynamics, transitioning from trench retreat to trench advance, triggered by the onset of a same-dip double subduction system (SDSS) on a plate capable of transmitting slab pull between trenches. This model illustrates the tectonic effects of this process, highlighting plate margin compression occurring away from the SDSS. For simplicity, the upper-right panel depicts a fixed upper plate. The conceptual framework builds upon the established understanding of SDSS dynamics^{2,3,5,6,13–15} and our own results on the tectonic effects on near by plate margins. **b** Proposal of a double subduction-induced orogenic event associated to the Neotethys SDSS system showing global^{14,71} and regional⁷⁴ plate kinematic reconstructions for the late Early to Late Cretaceous.